RL-BCM: Detailed Experiments and Reproducibility Report

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Companion report to "Reinforcement Learning for Betweenness Centrality Maximization via Graph Structure Modification" (CSoNet 2026)

September 2026

This report contains additional measured results and implementation details for the CSoNet 2026 paper. The paper is self-contained for its principal claims. All reported experiments were executed locally; unrun baselines are explicitly marked as such.

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1 Scope

Full Greedy is stronger than RL in all three synthetic-family means, and the learned myopic control is also stronger than the long-horizon model. The results support a learned heuristic with a measurable quality/cost profile, not a demonstrated planning advantage. All methods use the same relative-gain definition, with the initial BC as denominator.

The protocol uses held-out validation, deterministic SHA-256 dataset seeds, dropout-free PPO likelihoods, a remaining-budget feature, unordered symmetric edge scores, and absolute terminal-aligned reward. The exact task adds a fixed number of edges; at-most-budget behavior is audited separately because BC is not monotone under unrestricted insertion.

2 Frozen experimental design

Training uses 480 graphs, validation 36, and test 240. Synthetic configurations are ER with probabilities 0.1/0.2, BA with attachment 2/3, and WS with degree–rewiring pairs (4,0.1)/(6,0.2). Connectedness rejection is explicit. Train sizes are 20,30,40,50; test sizes are 30,50,80,100. Each test family has two configurations, four sizes, and ten graphs per cell. Targets come from the bottom quartile of positive BC. Saved graph hashes verify disjoint splits.

Five seeds (11,22,33,44,55) are trained for each of exact-BC, no-BC-input, and gamma-zero models. Every run executes 5,000 episodes of five decisions. All variants use three width-64 mean GraphSAGE layers, a 192–64–32–1 edge MLP, a 128–64–1 value head, and the same optimizer settings. Adam uses $3 \cdot 10^{-4}$; PPO uses clip 0.2, four epochs, minibatches of 32, 16 episodes per rollout, entropy 0.01, value coefficient 0.5, gradient clipping 0.5, and GAE lambda 0.95. Main gamma is one; the control changes only gamma to zero. The final checkpoint is used without test or validation selection.

3 Metric definitions and uncertainty

Primary outcomes are normalized endpoint-excluding final BC and its absolute increase. Relative increase is $(B_k - B_0)/B_0$ only for $B_0 > 10^{-12}$. The mean of relative increases is not a ratio of means and can be dominated by small initial values. We report median relative gain as a secondary metric.

For learned models the measurement matrix has five seed rows and graph columns. The crossed bootstrap independently samples seed indices and graph indices, using the same sampled graph columns for both members of a contrast. Graphs are stratified by configuration and size. We use 10,000 replicates with fixed bootstrap seed 92831. This estimates both observed training-run and test-graph variation; it does not incorporate new training corpora or new network families. For deterministic baselines, only graph columns vary. Original Random and global-sampled baselines have one predefined randomization per graph, so their within-graph randomization variance is not independently estimated.

Seed-level sign-flip tests use graph-averaged paired differences and all 32 sign assignments. They are two-sided, require a symmetric null for these differences, and have minimum attainable unadjusted $p = 0.0625$. Holm correction is applied across ER, BA, and WS for each specified comparator. A bootstrap interval excluding zero and a coarse sign-flip p-value need not agree: the two procedures use different sampling assumptions and resolution. We do not describe this discrepancy as proof of statistical significance.

4 Full primary and secondary metrics

Family	Method	B_0	B_k	$B_k - B_0$	Median rel. (%)
ER	RL-BCM	0.00747	0.08007	0.07260	620.8
ER	RL without BC	0.00747	0.07848	0.07101	622.1
ER	RL myopic	0.00747	0.08086	0.07339	648.2
ER	Full Greedy	0.00747	0.08355	0.07608	640.0
ER	Incident Greedy	0.00747	0.08355	0.07608	640.0
ER	Sampled Greedy (20)	0.00747	0.04004	0.03257	164.1
ER	Sampled Greedy (100)	0.00747	0.07022	0.06275	398.0
ER	Degree	0.00747	0.04560	0.03814	244.6
ER	Random	0.00747	0.00889	0.00142	-1.4
BA	RL-BCM	0.00203	0.06278	0.06075	3085.6
BA	RL without BC	0.00203	0.06412	0.06209	3162.2
BA	RL myopic	0.00203	0.06630	0.06427	3260.9
BA	Full Greedy	0.00203	0.07134	0.06930	3526.4
BA	Incident Greedy	0.00203	0.07134	0.06930	3526.4
BA	Sampled Greedy (20)	0.00203	0.02225	0.02022	759.9
BA	Sampled Greedy (100)	0.00203	0.05144	0.04940	2610.9
BA	Degree	0.00203	0.02369	0.02165	940.9
BA	Random	0.00203	0.00310	0.00107	-1.1
WS	RL-BCM	0.01263	0.26387	0.25124	1927.6
WS	RL without BC	0.01263	0.25919	0.24656	1953.8
WS	RL myopic	0.01263	0.26778	0.25515	1994.5
WS	Full Greedy	0.01263	0.28227	0.26964	2119.9
WS	Incident Greedy	0.01263	0.28227	0.26964	2119.9
WS	Sampled Greedy (20)	0.01263	0.12735	0.11472	733.5
WS	Sampled Greedy (100)	0.01263	0.23348	0.22085	1674.0
WS	Degree	0.01263	0.20811	0.19548	1534.7
WS	Random	0.01263	0.01654	0.00391	-4.2

4.1 All training-seed means

Table 2: Mean final BC per training seed on the same 80 graphs per family. SD is between the five seed means, not between graph instances.

Family / model	11	22	33	44	55	SD
ER / RL-BCM	0.0809	0.0803	0.0796	0.0793	0.0802	0.0006
ER / RL without BC	0.0790	0.0782	0.0784	0.0788	0.0780	0.0004
ER / RL myopic	0.0810	0.0808	0.0818	0.0794	0.0812	0.0009
BA / RL-BCM	0.0612	0.0641	0.0630	0.0622	0.0634	0.0011
BA / RL without BC	0.0620	0.0658	0.0632	0.0634	0.0663	0.0018
BA / RL myopic	0.0667	0.0661	0.0649	0.0671	0.0667	0.0009
WS / RL-BCM	0.2671	0.2691	0.2639	0.2581	0.2611	0.0044
WS / RL without BC	0.2498	0.2528	0.2702	0.2600	0.2631	0.0082
WS / RL myopic	0.2693	0.2574	0.2698	0.2727	0.2697	0.0060

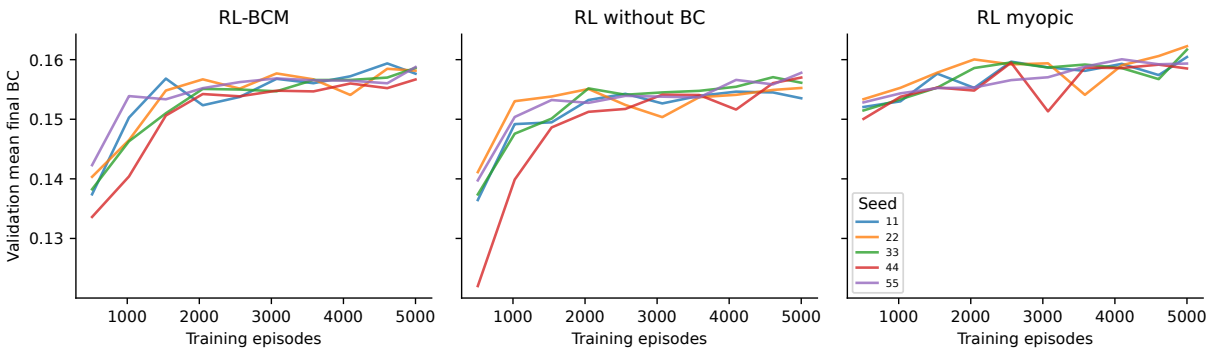


Figure 1: Held-out validation for all seeds; final checkpoints are used without selection.

4.2 Paired comparisons

Family	Contrast	Difference	95% CI	Win%	Tie%	p_H
ER	exact - FullGreedy	-0.00348	[-0.00479,-0.00228]	6.5	6.2	0.188
ER	exact - IncidentGreedy	-0.00348	[-0.00479,-0.00228]	6.5	6.2	0.188
ER	exact - Degree	+0.03446	[+0.02988,+0.03951]	100.0	0.0	0.188
ER	exact - no_bc	+0.00159	[+0.00064,+0.00267]	63.7	10.5	0.188
ER	exact - myopic	-0.00079	[-0.00184,+0.00015]	34.0	11.5	0.188
BA	exact - FullGreedy	-0.00856	[-0.00990,-0.00726]	2.5	2.8	0.188
BA	exact - IncidentGreedy	-0.00856	[-0.00990,-0.00726]	2.5	2.8	0.188
BA	exact - Degree	+0.03909	[+0.03643,+0.04189]	99.8	0.0	0.188
BA	exact - no_bc	-0.00134	[-0.00246,-0.00022]	35.2	12.2	0.188
BA	exact - myopic	-0.00352	[-0.00511,-0.00196]	20.2	11.8	0.188
WS	exact - FullGreedy	-0.01840	[-0.02243,-0.01446]	11.5	0.0	0.188
WS	exact - IncidentGreedy	-0.01840	[-0.02243,-0.01446]	11.5	0.0	0.188
WS	exact - Degree	+0.05576	[+0.04573,+0.06609]	95.5	0.0	0.188
WS	exact - no_bc	+0.00468	[-0.00409,+0.01399]	46.5	3.2	0.500
WS	exact - myopic	-0.00391	[-0.01116,+0.00488]	32.5	2.8	0.438

4.3 Breakdown by graph size

Family, n	Degree	SG20	SG100	Full G.	RL	No BC	Myopic
ER, 30	0.12742	0.11566	0.19457	0.21863	0.21020	0.20753	0.21191
ER, 50	0.03295	0.02755	0.05720	0.07053	0.06705	0.06508	0.06802
ER, 80	0.01331	0.01078	0.01813	0.02803	0.02677	0.02576	0.02706
ER, 100	0.00874	0.00615	0.01097	0.01699	0.01625	0.01554	0.01644
BA, 30	0.03592	0.05003	0.09769	0.10641	0.10211	0.09968	0.10316
BA, 50	0.02333	0.02472	0.05566	0.07624	0.07032	0.07041	0.07280
BA, 80	0.01870	0.00833	0.03003	0.05498	0.04324	0.04663	0.04822
BA, 100	0.01681	0.00592	0.02236	0.04771	0.03545	0.03976	0.04103
WS, 30	0.22493	0.17998	0.27979	0.28766	0.28142	0.28010	0.28217
WS, 50	0.21272	0.14489	0.25138	0.28792	0.27873	0.27588	0.28282

Family, n	Degree	SG20	SG100	Full G.	RL	No BC	Myopic
WS, 80	0.21265	0.11507	0.22262	0.28504	0.26407	0.25930	0.27049
WS, 100	0.18214	0.06944	0.18013	0.26845	0.23126	0.22147	0.23564

4.4 At-most-budget diagnostic

The following means select the best prefix of the stored five-step trajectory, including no modification. This is an oracle-assisted diagnostic, not a separately trained stopping policy. No-BC deployment would require additional BC evaluation to choose such a prefix.

Family	Method	Fixed budget	Best prefix
ER	Degree	0.045605	0.045605
ER	Full Greedy	0.083545	0.083545
ER	RL-BCM	0.080069	0.080069
ER	RL without BC	0.078477	0.078477
ER	RL myopic	0.080857	0.080857
BA	Degree	0.023688	0.023688
BA	Full Greedy	0.071335	0.071335
BA	RL-BCM	0.062780	0.062780
BA	RL without BC	0.064123	0.064123
BA	RL myopic	0.066304	0.066304
WS	Degree	0.208111	0.208111
WS	Full Greedy	0.282269	0.282269
WS	RL-BCM	0.263868	0.263868
WS	RL without BC	0.259187	0.259187
WS	RL myopic	0.267780	0.267780

5 Exact-optimum and budget diagnostics

On 60 predefined eight-vertex graphs with $k = 2$, exhaustive enumeration finds that Full Greedy is suboptimal in 6 cases. Its mean final BC is 0.1583, compared with the exact optimum mean 0.1594. RL-BCM has mean final BC 0.1188 and reaches the optimum in 46.3% of its 300 seed–graph outcomes. RL without BC has mean final BC 0.1236 and reaches the optimum in 54.3% of its 300 seed–graph outcomes. RL myopic has mean final BC 0.1391 and reaches the optimum in 59.7% of its 300 seed–graph outcomes. These failures prevent interpreting the MDP formulation as evidence that RL reliably escapes greedy traps.

Across 60 size-50 test graphs (20 per family), the same fixed policy is tested at $k = 1, 3, 10$. At $k = 1$, mean final BC is 0.0348 for RL and 0.0415 for Full Greedy. At $k = 3$, mean final BC is 0.0900 for RL and 0.0966 for Full Greedy. At $k = 10$, mean final BC is 0.2350 for RL and 0.2408 for Full Greedy. The optimal benchmark enumerates all unordered two-edge sets. With q absent edges this means $\binom{q}{2}$ final graphs. Enumeration of sets is sufficient because the final graph is order-independent, even though greedy's choices depend on intermediate graphs. No global-optimality claim is made for greedy.

5.1 Recorded counterexample

Vertices are 0 through 7, target 6, and edges are $\{02, 03, 04, 16, 47, 56, 57, 67\}$. The greedy trajectory adds 06 and 26 and ends at 0.5238095238. The optimal set adds 26 and 36 and ends at 0.5317460317. All 190 missing-edge pairs were evaluated. Its complete graph and target record is `tiny_ER_0.3_8_4`. This is a verified greedy failure, not a claim that RL solves this particular graph.

5.2 Budget transfer by family

The model always trained at budget five. Initial budget ratios equal one at every test horizon, so that feature alone does not reveal the absolute horizon at the first step. Transfer to budgets one, three, and ten is therefore a stronger distribution shift than merely changing remaining steps in a fixed five-step episode. Dedicated horizon-conditioned retraining would be a separate experiment and is not claimed here.

Family	k	Degree	Incident G.	Full G.	RL
ER	1	0.01067	0.02029	0.02029	0.01897
ER	3	0.02086	0.04499	0.04499	0.04198
ER	10	0.07312	0.13708	0.13708	0.13363
BA	1	0.00370	0.01505	0.01505	0.01192
BA	3	0.01024	0.04338	0.04338	0.03981
BA	10	0.08193	0.16200	0.16200	0.15522
WS	1	0.06539	0.08905	0.08905	0.07353
WS	3	0.14309	0.20129	0.20129	0.18824
WS	10	0.34909	0.42344	0.42344	0.41618

6 Real-network preprocessing and structure

SNAP email-Eu-core is converted to an undirected simple graph. Both datasets have self-loops removed before retaining their largest connected component. The original email directed edge count must not be compared directly to the processed undirected count. GrQc's published LCC edge count also includes edges removed by the loop-cleaning step. Source file checksums and vertex-ID maps are provided with the graph records.

BFS and random-walk samplers select vertices, then retain the induced graph. The same source network can contribute overlapping samples. No confidence interval across these samples is interpreted as independent-network uncertainty. Degree-biased visitation and locality bias remain for random walks and BFS, respectively; neither is called a representative uniform network sample.

Table 7: Mean final BC on three targets/subgraphs per row. Incident SG20/SG100 and learned models average five seeds; these are not independent source networks. Dashes denote unrun methods. RL quality uses GPU inference; no cross-device speedup is inferred.

Network/sample	Degree	ISG20	ISG100	Inc. G.	RL	No BC
Email BFS 80	0.0047	0.0132	0.0137	0.0137	0.0121	0.0115
Email RW 80	0.0020	0.0287	0.0337	0.0337	0.0304	0.0268
Email BFS 256	0.0004	0.0062	0.0089	0.0096	0.0072	0.0068
Email RW 256	0.0008	0.0042	0.0062	0.0067	0.0050	0.0050
Email LCC 986	0.0002	0.0031	0.0058	–	0.0061	0.0053
GrQc BFS 80	0.2838	0.2808	0.3168	0.3168	0.2720	0.2518
GrQc RW 80	0.2463	0.4687	0.5032	0.5032	0.3520	0.3651
GrQc BFS 256	0.0608	0.0700	0.1196	0.1386	0.0398	0.0608
GrQc RW 256	0.0046	0.2811	0.3234	0.3415	0.1553	0.1543
GrQc LCC 4158	0.0042	0.0195	0.0297	–	0.0072	0.0092

The incident-sampled controls extend the stronger comparison to both full LCCs. On email-Eu-core, Incident SG20/SG100 attain 0.0031/0.0058, compared with 0.0061 for RL. On ca-GrQc, Incident SG20/SG100 attain 0.0195/0.0297, compared with 0.0072 for RL. Transfer also has substantial failures: on GrQc BFS-256, RL attains 0.0398 versus 0.1386 for Incident Greedy. Global SG20 on Email LCC is 1.865×10^{-5}, not zero. Full Greedy remains unrun above 80 vertices on these networks.

6.1 Measured structural properties

Network/sample	n	m	Mean degree	Density	Clustering	Transitivity
Email LCC 0	986	16064	32.58	0.0331	0.4071	0.2674
Email BFS 0	80	1023	25.57	0.3237	0.6672	0.5143
Email BFS 1	80	517	12.93	0.1636	0.5187	0.4233
Email BFS 2	80	947	23.68	0.2997	0.5706	0.5049
Email BFS 0	256	4409	34.45	0.1351	0.4834	0.3736
Email BFS 1	256	4578	35.77	0.1403	0.5241	0.3834
Email BFS 2	256	3952	30.88	0.1211	0.4892	0.3666
Email RW 0	80	520	13.00	0.1646	0.4812	0.3836
Email RW 1	80	638	15.95	0.2019	0.4699	0.3976
Email RW 2	80	420	10.50	0.1329	0.3965	0.3267
Email RW 0	256	4071	31.80	0.1247	0.4719	0.3665
Email RW 1	256	3848	30.06	0.1179	0.4661	0.3548
Email RW 2	256	4344	33.94	0.1331	0.4843	0.3842
GrQc LCC 0	4158	13422	6.46	0.0016	0.5569	0.6289
GrQc BFS 0	80	201	5.03	0.0636	0.7156	0.3255
GrQc BFS 1	80	537	13.43	0.1699	0.7709	0.8839
GrQc BFS 2	80	172	4.30	0.0544	0.5291	0.2414
GrQc BFS 0	256	571	4.46	0.0175	0.5530	0.2426
GrQc BFS 1	256	724	5.66	0.0222	0.6268	0.4683
GrQc BFS 2	256	821	6.41	0.0252	0.5547	0.6272
GrQc RW 0	80	163	4.08	0.0516	0.6429	0.6524

Network/sample	n	m	Mean degree	Density	Clustering	Transitivity
GrQc RW 1	80	162	4.05	0.0513	0.5094	0.4922
GrQc RW 2	80	221	5.53	0.0699	0.5654	0.6949
GrQc RW 0	256	1711	13.37	0.0524	0.7100	0.8472
GrQc RW 1	256	1286	10.05	0.0394	0.5542	0.8273
GrQc RW 2	256	1341	10.48	0.0411	0.6507	0.8349

7 Cost measurements and implementation

Table 9: Matched quality/cost at $n = 256$: mean final BC (top) and median CPU decision seconds (bottom) on five graphs per family, three timing repetitions, model seed 11. One thread; audit BC, loading, compilation and training excluded. ISG20 samples incident edges. All sizes/methods are in the report.

Family	Degree	ISG20	Inc. G.	RL	No BC	Myopic
ER	0.0026 0.0055 s	0.0036 0.1042 s	0.0038 0.9696 s	0.0037 0.3118 s	0.0037 0.2690 s	0.0038 0.3051 s
BA	0.0236 0.0053 s	0.0259 0.0873 s	0.0409 0.9950 s	0.0119 0.2919 s	0.0144 0.2823 s	0.0211 0.2969 s
WS	0.2518 0.0057 s	0.3016 0.0905 s	0.3463 0.9997 s	0.2636 0.3068 s	0.2198 0.2889 s	0.2953 0.3153 s

Table 9 pairs quality and time on identical graphs; the main manuscript figure shows BA-256. Another measured method dominates the main policy in 8/9 family–size cells when comparing mean BC and median time. For ER-100, RL attains BC 0.0208 in 0.0479 s, versus 0.0217 in 0.0590 s for Incident Greedy: a conditional tradeoff, not superiority. For BA-256, ISG20 has both higher BC (0.0259 versus 0.0119) and lower latency (0.0873 versus 0.2919 s). These are descriptive comparisons on five graphs per cell and one model seed. The five main training runs cost 37.1 cumulative process-minutes, including validation; overlapping runs prevent interpreting this sum as wall-clock duration. Any training-cost amortization requires repeated uses, positive latency savings, and acceptable quality.

Training elapsed times below are process durations measured while some jobs overlapped. They include validation evaluations. Summing them is not total wall-clock time and is not energy consumption.

Variant	Seed	Episodes	PPO updates	Elapsed (s)
exact	11	5000	313	456.08
exact	22	5000	313	489.16
exact	33	5000	313	471.64
exact	44	5000	313	408.88
exact	55	5000	313	403.13
no_bc	11	5000	313	371.97
no_bc	22	5000	313	394.64
no_bc	33	5000	313	380.96
no_bc	44	5000	313	392.27
no_bc	55	5000	313	392.20
myopic	11	5000	313	374.61

Variant	Seed	Episodes	PPO updates	Elapsed (s)
myopic	22	5000	313	396.57
myopic	33	5000	313	380.47
myopic	44	5000	313	393.40
myopic	55	5000	313	395.05

7.1 Exact insertion oracle

For an inserted edge uv, new shortest paths between s, t either avoid that edge or traverse it in one of its two orientations. Candidate distances are d_{st}, $d_{su} + 1 + d_{vt}$, and $d_{sv} + 1 + d_{ut}$. Counts for all tied minimizers are added. An alleged minimizing walk with repeated vertices would contain a removable positive-length cycle, contradicting minimality; products of the original path counts are therefore valid for the minimizing cases. Disconnected pairs have zero counts and do not contribute. Candidate target BC is then obtained from all updated pair distances/counts, including distances to the target. The tables are rebuilt after each committed edge, so this does not reuse obsolete marginals. Tie tolerance is 10^{-12}.

7.2 Source-level checks and artifact audit

The automated suite compares Brandes to NetworkX; compares all candidate insertions to independent full BC recomputation; checks exhaustive greedy's stepwise maximum; tests clustering features and the absent BC channel; and tests score equivariance under vertex relabeling. A post-hoc audit checks exact row counts, unique method/seed/graph keys, valid new edges, trajectory lengths, metric identities, checkpoint completion, and replays final BC. Deterministically selected outcomes are independently recomputed with NetworkX.

The files named `baselines.jsonl`, `real_baselines.jsonl`, and `real_rl.jsonl` are explicitly excluded development records. Final quality data use the `v2` baseline files and `real_rl_cuda.jsonl`. See `EXCLUDED_RUNS.md`.

8 Reproduction

Run from the repository root with the recorded Python environment. The train stage skips completed runs. Remove or move a run directory only when deliberately starting a fresh reproduction, and keep the original artifact separate.

```
python -m unittest discover -s tests -v
python -m experiments.revision_run prepare
python -m experiments.revision_suite --workers 2
python -m experiments.revision_run baselines
python -m experiments.revision_run evaluate
python -m experiments.revision_extra tiny
python -m experiments.revision_extra tiny_rl
python -m experiments.revision_extra budget
python -m experiments.revision_extra prepare_real
python -m experiments.revision_extra real_baselines
python -m experiments.revision_extra real_rl --device cuda
python -m experiments.revision_timing
python -m experiments.revision_report
python -m experiments.followup_experiments prepare
```

```
python -m experiments.followup_experiments repair_tiny
python -m experiments.followup_experiments annotate_budgets
python -m experiments.followup_experiments synthetic
python -m experiments.followup_experiments real
python -m experiments.followup_experiments timing
python -m experiments.revision_audit
python -m experiments.followup_experiments audit
python -m experiments.followup_report
python -m experiments.revision_supplement
```

CPU execution can replace GPU execution for real-network quality evaluation, but the output filename and reporting input must be changed consistently. GPU and CPU floating-point scoring can break near ties differently; the deployed edge trajectory is always retained and scored with the exact CPU objective. Timing requires matching hardware and exclusion of concurrent training or evaluation load. Dataset source files are local copies of the public SNAP edge lists; no new account or paid compute is required.

9 Incident-sampled controls and matched timing

This follow-up was specified after the initial evaluation. It retains all 15 checkpoints without further tuning. Each incident-sampled baseline has five random seeds (101, 202, 303, 404, 505) on 240 synthetic and 30 real instances: 2,700 added quality outcomes. The timing set contains 45 new graphs (five per family and size 50, 100, 256), with three repetitions, one thread, and model seed 11. Technical repetitions do not increase the number of independent graphs.

Sampling within the incident action set changes the comparison. Incident SG20 reaches 0.0817, 0.0669, 0.2728 mean final BC on ER, BA, and WS. Incident SG100 evaluates every incident candidate when $n \leq 100$ and therefore reproduces Incident Greedy on this test set. These controls distinguish a limited search budget from spending that budget on a much larger action space.

The pooled optimum rate hides ER/BA/WS rates of 22%, 22%, and 95%. On the six instances where Full Greedy is suboptimal, the main policy beats greedy and reaches the optimum in only 1/30 seed–graph outcomes. Six tiny instances cannot accommodate two incident additions; the artifact now marks Incident Greedy infeasible there and reports no fixed-budget final score.

Table 9 pairs quality and time on identical graphs; the main manuscript figure shows the BA-256 tradeoff. Incident SG20 obtains BC 0.0259 in 0.0873 s. Incident Greedy obtains BC 0.0409 in 0.9950 s. RL-BCM obtains BC 0.0119 in 0.2919 s. No BC obtains BC 0.0144 in 0.2823 s. A latency saving is useful only if the associated quality is acceptable. The five main-model training runs cost 37.1 cumulative process-minutes (101.7 for all 15 runs), including validation; overlapping processes prevent interpreting these sums as wall-clock duration. Using seed 11's 456.1 s training duration and the BA-100 median saving over Incident Greedy gives an illustrative break-even of 35,980 uses, before accounting for unequal quality. This is a conditional calculation, not a universal deployment threshold.

The incident candidate generator directly lists target non-neighbors in lexicographic order. It avoids constructing all nonedges for incident methods; earlier latency records remain archived and are not pooled with this new benchmark. Infeasible fixed-budget requests return a null final score, a status flag, and the requested/used budget. No unrestricted fallback is silently substituted.

Incident monotonicity follows directly from the path-fraction definition. For endpoints distinct from the target, every new path using an inserted target-incident edge passes through the target. If new shortest paths are shorter, the contribution becomes one; if they tie existing shortest paths, the fraction changes from a/b to $(a + c)/(b + c)$ with $a \leq b$; otherwise it is unchanged. Previously

disconnected pairs cannot decrease their zero contribution. Summing with a fixed normalization proves nondecrease, not global optimality of an incident restriction.

Family	Method	n	Mean final BC	Median seconds
ER	Degree	50	0.052788	0.000640
ER	Full Greedy	50	0.105747	0.162499
ER	Incident Greedy	50	0.105747	0.007190
ER	Global SG20	50	0.049844	0.003936
ER	Incident SG100	50	0.105747	0.007396
ER	Incident SG20	50	0.103086	0.003845
ER	RL-BCM	50	0.100877	0.009666
ER	Myopic	50	0.103485	0.009658
ER	No BC	50	0.100047	0.008851
ER	Degree	100	0.011312	0.001230
ER	Full Greedy	100	0.021698	2.709064
ER	Incident Greedy	100	0.021698	0.059032
ER	Global SG20	100	0.006985	0.015452
ER	Incident SG100	100	0.021698	0.058554
ER	Incident SG20	100	0.019902	0.015157
ER	RL-BCM	100	0.020802	0.047899
ER	Myopic	100	0.021384	0.048177
ER	No BC	100	0.020734	0.044557
ER	Degree	256	0.002580	0.005540
ER	Incident Greedy	256	0.003825	0.969571
ER	Global SG20	256	0.001955	0.106297
ER	Incident SG100	256	0.003811	0.418815
ER	Incident SG20	256	0.003611	0.104167
ER	RL-BCM	256	0.003713	0.311830
ER	Myopic	256	0.003757	0.305093
ER	No BC	256	0.003663	0.269020
BA	Degree	50	0.036228	0.000611
BA	Full Greedy	50	0.094487	0.169474
BA	Incident Greedy	50	0.094487	0.007349
BA	Global SG20	50	0.019957	0.003948
BA	Incident SG100	50	0.094487	0.007434
BA	Incident SG20	50	0.086525	0.003740
BA	RL-BCM	50	0.077697	0.009927
BA	Myopic	50	0.085731	0.009598
BA	No BC	50	0.075406	0.008714
BA	Degree	100	0.036126	0.001325
BA	Full Greedy	100	0.069869	2.861243
BA	Incident Greedy	100	0.069869	0.059216
BA	Global SG20	100	0.009936	0.014033
BA	Incident SG100	100	0.069869	0.059894
BA	Incident SG20	100	0.063336	0.013588
BA	RL-BCM	100	0.048055	0.046539
BA	Myopic	100	0.059275	0.050862

Family	Method	n	Mean final BC	Median seconds
BA	No BC	100	0.047887	0.046297
BA	Degree	256	0.023551	0.005337
BA	Incident Greedy	256	0.040882	0.995029
BA	Global SG20	256	0.000776	0.089492
BA	Incident SG100	256	0.035404	0.398642
BA	Incident SG20	256	0.025893	0.087267
BA	RL-BCM	256	0.011884	0.291900
BA	Myopic	256	0.021080	0.296916
BA	No BC	256	0.014447	0.282307
WS	Degree	50	0.325878	0.000598
WS	Full Greedy	50	0.414083	0.167057
WS	Incident Greedy	50	0.414083	0.006956
WS	Global SG20	50	0.160054	0.003822
WS	Incident SG100	50	0.414083	0.007232
WS	Incident SG20	50	0.396988	0.003721
WS	RL-BCM	50	0.402955	0.009080
WS	Myopic	50	0.408401	0.009322
WS	No BC	50	0.370992	0.008793
WS	Degree	100	0.311090	0.001255
WS	Full Greedy	100	0.438549	2.915208
WS	Incident Greedy	100	0.438549	0.059081
WS	Global SG20	100	0.143017	0.014064
WS	Incident SG100	100	0.438549	0.059103
WS	Incident SG20	100	0.406165	0.013720
WS	RL-BCM	100	0.395825	0.051661
WS	Myopic	100	0.391993	0.049709
WS	No BC	100	0.355062	0.046353
WS	Degree	256	0.251800	0.005696
WS	Incident Greedy	256	0.346315	0.999685
WS	Global SG20	256	0.079504	0.090090
WS	Incident SG100	256	0.323468	0.413864
WS	Incident SG20	256	0.301575	0.090542
WS	RL-BCM	256	0.263582	0.306776
WS	Myopic	256	0.295342	0.315332
WS	No BC	256	0.219844	0.288937

The matched mean-quality/median-time comparison finds the main policy dominated by another measured method in 8 of nine family–size cells. This is descriptive evidence on five graphs per cell, not a significance or universal dominance claim. Full records retain all seed and graph outcomes.

10 Remaining limitations

There is no evidence here that longer-horizon PPO beats a matched immediate reward policy. Exhaustive greedy is unrun on large LCCs. Five training seeds give only coarse inferential resolution. Low-BC target selection, a fixed training corpus, two real datasets, potential subgraph overlap, one

original global-sampled baseline realization per graph (the added incident controls have five), dense adjacency, and quadratic pair scoring limit the scope of conclusions. The report makes no universal optimization, scalability, reproducibility-across-hardware, or state-of-the-art claim.